12.3 What Does "Random" Mean?

WORKSHEET

The question

STEP III 2022, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If the angle AOB is θ, what is the length of the chord AB? (Drop a perpendicular from O.)

STAGE 2 A random point inside

Ask yourself What is $\mathrm{P}(R \le r)$ for $0 \le r \le a$? Differentiate to get the density.

Ask yourself A chord through C makes angle t with OC. How far is the chord from O?

STAGE 3 Averaging over C

Ask yourself Factorise $1 - \cos^3 t$ and $\sin^2 t = 1 - \cos^2 t$. What cancels?

Try this next

1. **Bertrand's probabilities.** Verify the three probabilities $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ above.
2. **The other mean.** Show that if the chord's midpoint is chosen uniformly in the disc (the right-hand picture), the expected length is $\frac{4a}{3}$. (*Hint:* this is $L(\frac{\pi}{2})$. Why?)
3. **Random distance.** If the distance of the chord from O is uniform on $[0, a]$, show that the expected length is $\frac{\pi a}{2}$.
4. **Three methods, three distributions.** For each of the three methods, find the probability density of the chord's length ℓ on $[0, 2a]$.

Further reading

- **Edwin Jaynes, "The well-posed problem"**, *Foundations of Physics* 3 (1973). The invariance argument, readable and provocative.
- **Gábor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel). Bertrand's paradox and many others.
- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press). Integral geometry, for later.

The full guided solution, and what the question leads to at university, is in Section 12.3:
https://beyondstep.keithmcnulty.org/ch12/bertrand.html